

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{yz} \\ 0 & T_{yz} & 0 \end{bmatrix}$$

$$T_{yz} = \mathbb{T}_{yz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{yz} \\ 0 & -T_{yz} & 0 \end{bmatrix}$$

$$T_{yz} = \mathbb{M}_x^{(1)}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xyz} & 0 & 0 \\ 0 & 0 & 0 & T_{yyy} & 0 & 0 \\ 0 & 0 & 0 & T_{zzy} & 0 & 0 \\ T_{xyz} & T_{yyy} & T_{zzy} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxy} \\ 0 & 0 & 0 & 0 & T_{zxy} & 0 \end{bmatrix}$$

$$T_{xyz} = 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)}$$

$$T_{yyy} = \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)}$$

$$T_{zzy} = -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)}$$

$$T_{zxy} = 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xyz} & 0 & 0 \\ 0 & 0 & 0 & T_{yyy} & 0 & 0 \\ 0 & 0 & 0 & T_{zzy} & 0 & 0 \\ -T_{xyz} & -T_{yyy} & -T_{zzy} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxy} \\ 0 & 0 & 0 & 0 & -T_{zxy} & 0 \end{bmatrix}$$

$$T_{xyz} = -4\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{yyy} = 2\mathbb{M}_x^{(1)} - 2\mathbb{M}_x^{\alpha(1)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{zzy} = -2\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{zxy} = -\mathbb{M}_x^{(1)} - 4\mathbb{M}_x^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zxy} \\ 0 & T_{zxy} & 0 \end{bmatrix}$$

$$T_{zxy} = 2\mathbb{T}_{yz}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zxy} \\ 0 & -T_{zxy} & 0 \end{bmatrix}$$

$$T_{zxxy} = -\mathbb{M}_x^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & 0 & 0 \\ T_{yyyz} & 0 & 0 \\ T_{zzyz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{zxxy} \\ 0 & T_{xyzx} & 0 \end{bmatrix}$$

$$T_{xxyz} = \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)}$$

$$T_{yyyy} = \mathbb{M}_x^{(3)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} - 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zzyz} = \mathbb{M}_x^{(3)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zxxy} = \mathbb{M}_x^{(4)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{yz}^{(5)}$$

$$T_{xyzx} = \mathbb{M}_x^{(4)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{yz}^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxxy} \\ 0 & 0 & 0 & 0 & T_{xyzx} & 0 \end{bmatrix}$$

$$T_{yzxx} = \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)}$$

$$T_{yzyy} = \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)}$$

$$T_{yzzz} = \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)}$$

$$T_{zxxy} = \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + \mathbb{T}_{yz}^{(6)}$$

$$T_{xyzx} = \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - \mathbb{T}_{yz}^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 & T_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & T_{yyyz} & 0 & 0 & T_{yyzy} & 0 & 0 \\ 0 & 0 & 0 & T_{zzyz} & 0 & 0 & T_{zzzy} & 0 & 0 \\ T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxxy} & 0 & 0 & T_{zxyx} \\ 0 & 0 & 0 & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} & 0 \end{bmatrix}$$

$$T_{xxyz} = \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{xxzy} = -\mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{(4)} - 2\mathbb{M}_x^{\alpha(2)} - 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)}$$

$$T_{yyyy} = 2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$

$$T_{yyzy} = 2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} - 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zzyz} = -2\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)}$$

$$T_{zzzy} = -2\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)}$$

$$T_{yzxx} = 4\mathbb{M}_x^{\beta(1)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} - \mathbb{T}_{yz}^{(3)}$$

$$T_{yzyy} = -2\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)}$$

$$T_{yzzz} = 2\mathbb{M}_x^{(1)} - 2\mathbb{M}_x^{\alpha(1)} - 2\mathbb{M}_x^{\beta(1)} - \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)}$$

$$T_{zxxy} = -\mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}$$

$$T_{zxyx} = -\mathbb{M}_x^{(1)} - \mathbb{M}_x^{(4)} - 4\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(5)}$$

$$T_{xyzx} = \mathbb{M}_x^{(1)} + \mathbb{M}_x^{(4)} + 4\mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(5)}$$

$$T_{xyxz} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{(4)} + 4\mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{zxxxy} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} \end{bmatrix}$$

$$\begin{aligned} T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\ T_{yzyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\ T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\ T_{zxxxy} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\ T_{zxyx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\ T_{xyzx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \\ T_{xyxz} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{yyyy} \\ 0 & T_{zzzy} & 0 \\ 0 & T_{yyzy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{xxyz} \\ 0 & 0 & T_{yzzz} \\ 0 & T_{zxxy} & 0 \\ 0 & 0 & 0 \\ -\frac{T_{yyyy}}{2} + \frac{T_{yyzy}}{2} + T_{zxxy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} T_{yyyy} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\ T_{zzzy} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\ T_{yyzy} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)} \\ T_{xxyz} &= 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)} \\ T_{yzzz} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)} \\ T_{zxxy} &= 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)} \end{aligned}$$